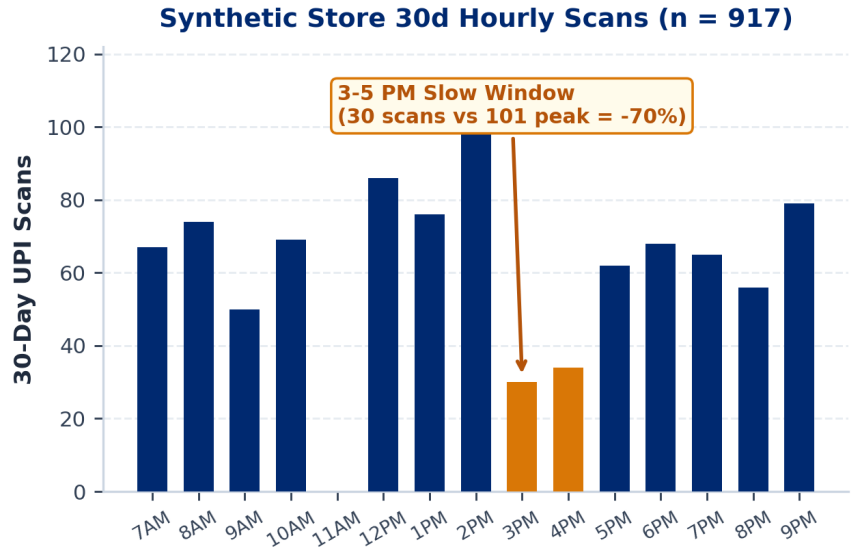
1.64 Cr Merchants Pay for Subscriptions; Retention Stays Manual

- **Reported Paytm Scale:** 1.64 Cr subscription merchants (+27L YoY), Rs 7.1L Cr GMV, Rs 2,648 Cr revenue (Reported: Paytm Q2 FY27 release, 26 Oct 2026).
- **Silent Repeat Churn:** Small shops lose an estimated 25% to 35% of repeat buyers yearly without customer-level alerts (Assumption).
- **Dashboard Friction:** Single-counter operators work 12+ hr shifts; SMB BI dashboards see <8% weekly usage (Assumption).
- **Slow Hours & Rush Prep:** In our synthetic Bengaluru ledger, 3 to 5 PM scans fall 70% below peak while weekend rushes risk stockouts.



PaytmAI VyaparVoice (Track 1)

8:30 PM closing-time voice retention & ops copilot for Soundbox + Paytm for Business.

Team SemiTechnical | Kevin Joseph

One 8:30 PM Voice Action, Governed by Deterministic SQL Rules

- **Plain Voice & Ergonomics:** EN author-checked, native review pending; KN/HI draft; Repeat, 0.78x Slower, Question Chips & Why? drawers.
- **Synthetic Store Slice:** Sri Krishna Darshini (240 payers, 1,613 scans); flags 48 regulars silent 16 to 38d.
- **Policy & 80/20 Split:** Filters to 40 DPDP & 14d-cooldown clear buyers: 32 nudged vs 8 silent control.
- **Cap Editor & Stop-Loss:** All 32 get Rs 25 pass; Cap Editor (Rs 50-2,500), 2-step voice confirm & 1-tap Stop-Loss protect pocket.
- **Zero LLM Math:** SQLite computes every rupee; per-visitor cloud sandboxes block paid API calls.

TRY LIVE SANDBOX DEMO

https://paytmai-vyaparvoice.onrender.com

Cloud Safeguards Active:
Isolated SQLite per visitor (30m TTL, 64-session LRU, 2MB/32MB disk caps).
Zero paid API calls on public URL.

Deterministic Core, Cap Controls & Scorecard Built; Rail Hooks Scoped

- **[BUILT] ITT Stats, Cap Editor & Stop-Loss:** Fisher/Wilson/power math, Cap Editor (Rs 50-2,500), 2-step voice confirm, atomic Stop-Loss, What-If & Guardrail Lab.
- **[BUILT] Bayesian Scorecard & Copy Ergonomics:** Plain verdict, Beta(1,1) profit posterior, 6x4 margin-fee table, Evidence Meter, Repeat/0.78x Slower, banned-jargon lint.
- **[BUILT KEYLESS / PENDING LIVE] Sarvam & Caps:** Keyless Bulbul TTS + Saaras STT adapter, mock-server selftest, SHA-256 audio cache, 15%/15% (30% max) credit cap.
- **[PRODUCTION PLAN] Paytm Ecosystem Hooks:** Paytm QR dynamic coupon settlement, 4G Soundbox push audio, WhatsApp Cloud supplier POs, Postgres/Redis.

6-Rule Guardrail Stress Lab

Blocks prompt injection, >22% discounts, unconsented blasts, and holdout removal.

Atomic Cap & Stop-Loss Lock

SQLite write locking enforces hard rupee stops and instant 1-tap campaign pause.

Bounded Visitor Sandboxes

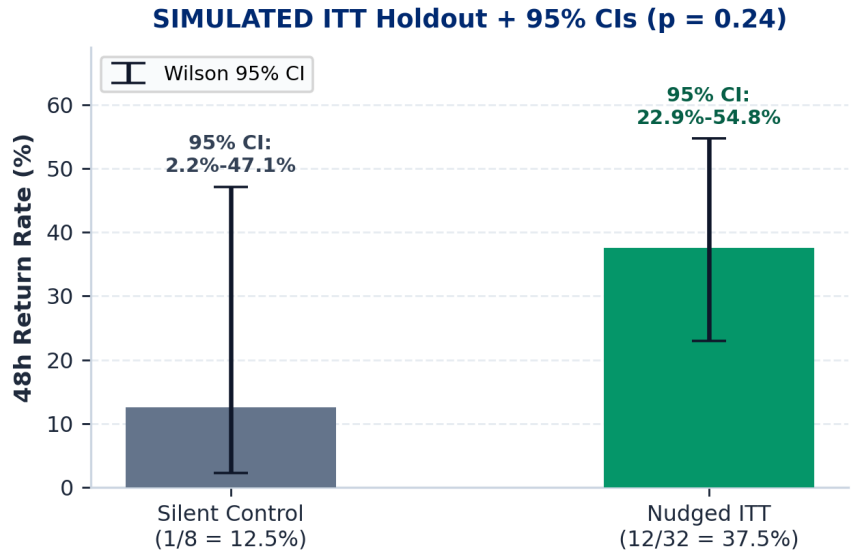
Per-visitor SQLite files enforce 30-min TTL, 64-session LRU eviction, and 32 MB disk cap.

PaytmAI VyaparVoice | Team SemiTechnical (Kevin Joseph)

Intention-to-Treat Holdout, Bayesian Posterior & Net Profit

- **Simulated ITT Return:** 12/32 nudged (37.5%) vs 1/8 control (12.5%); Fisher p = 0.24; simulated 12 equals the Rs 300 cap (0 refused).
- **Bayesian Posterior (n=40):** Beta(1,1) prior: 88.8% P(lift>0), 67.0% P(pre-fee profit>0), 52.1% P(post-fee profit>0); proves 50-shop pool is needed.
- **Sales vs Subsidy:** 8 net visits = Rs 1,740 sales (5.8x sales/promo); 4 organic returners absorb Rs 100 cashback.
- **Merchant Net Profit:** At 30% margin: Rs 522 GP (1.74x); +Rs 222 pre-fee (+74%), +Rs 91.50 after Rs 130.50 fee.
- **5% Fee Base:** Rs 130.50 = 5% of all Rs 2,610 redeemed bills (12 bills, including organic returners).

Intention-to-Treat (ITT) Definition: Compares all 32 assigned nudged buyers vs all 8 assigned silent controls regardless of redemption or cap stops, preventing selection bias and keeping lift invariant if the cap binds.



CAP & BREAK-EVEN DISCLOSURE (SIMULATED)

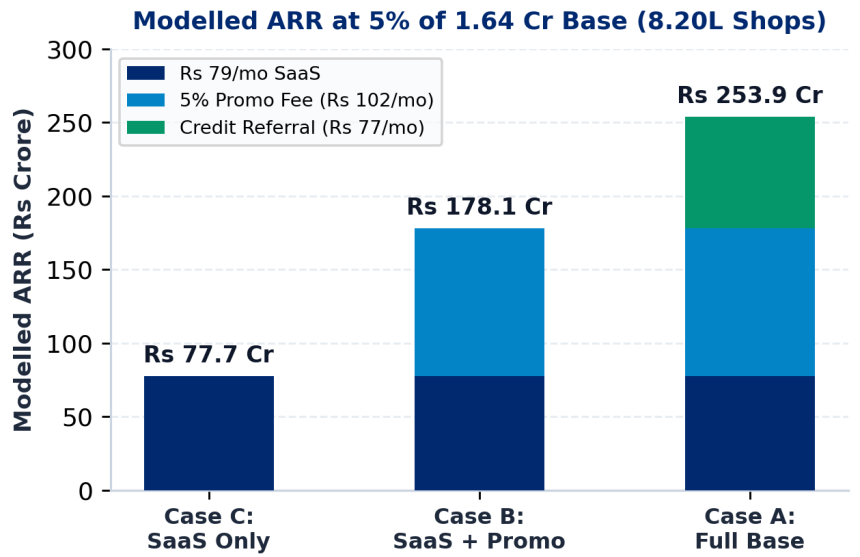
Simulated 12 returners = 12 x Rs 25 = Rs 300 hard spend stop (42.2% overrun risk at p=37.5% triggers >10% alert; suggests Rs 400 cap).

Break-even margin: 17.2% pre-fee | 24.7% post-fee.

Break-even lift (30% margin): +7.8 pp pre | +15.3 pp post.

Three Revenue Cases (8.20L Shops) & 50-Shop Pooled Pilot

- **Case C (SaaS Only):** Rs 79/mo Soundbox add-on x 8.20L shops = Rs 77.7 Cr ARR (zero promo/credit fees).
- **Case B (Core Excl. Credit):** +Rs 102/mo steady promo fee (~0.78 waves/mo, Rs 2,040 redeemed bills) = Rs 178.1 Cr ARR.
- **Case A (Full Base):** +Rs 77/mo credit referral (~7% loan pen. Assumption) = Rs 253.9 Cr ARR.
- **Unit COGS & Margin:** Rs 48/mo cached (Rs 5 AI + Rs 43 msg; 81.4% margin) vs Rs 71.80 uncached TTS (72.2%).
- **Pooled 50-Shop Pilot:** 50 shops x 1 wave (~400C + ~1,600T, stratified by shop): 83.3% power at +6 pp, 97.2% at +8 pp.



Why Pool 50 Shops (Not 4 Waves/Shop): 14d cooldown & ~0.78 waves/mo replenishment preclude 4 weekly waves in 1 shop; pooling 50x1 gives 50.3%/68.8%/83.3%/92.5%/97.2%/99.8% power at +4/+5/+6/+7/+8/+10 pp.

Q2 FY27 BASE & FEE RECONCILIATION

5% of 1.64 Cr subscription merchants (Reported: Paytm Q2 FY27, 26 Oct 2026) = 8.20L shops.

1 backlog wave = Rs 130.50 fee; steady state (~0.78 waves/mo) = Rs 102/mo.

APPENDIX A1 | TRACK 1: MERCHANT GROWTH AI (STAGE 2 VERIFIED - DECK V3)

PaytmAI VyaparVoice

Closing-Time Voice Growth, Operations & Retention Copilot

Built for Paytm's 1.64 Cr subscription-paying merchant deployments (Reported: Paytm Q2 FY27 release, 26 Oct 2026). Includes Cap Editor, 2-step voice confirmation, Stop-Loss, and Bayesian scorecard.

01 : Voice + Teach-Back

Repeat, 0.78x Slower, 5 Question Chips & Why? drawers

02 : Cap Editor & Stop-Loss

Rs 50-2,500 bounds, 2-step voice confirm & 1-tap halt

03 : Bayesian Scorecard

Beta(1,1) profit posterior + 6x4 margin-fee table

TEAM: SemiTechnical | Member: Kevin Joseph | Live: paytmai-vyaparvoice.onrender.com

Deck v3 Status: Stage 2 verified (stage2-ok); Cap Editor (A12) and Bayesian/Sensitivity Scorecard (A13) added.

Problem Details, Primary Paytm Q2 FY27 Sources & Synthetic Persona

1. Silent Repeat Churn (25% to 35% Yearly - ASSUMPTION)

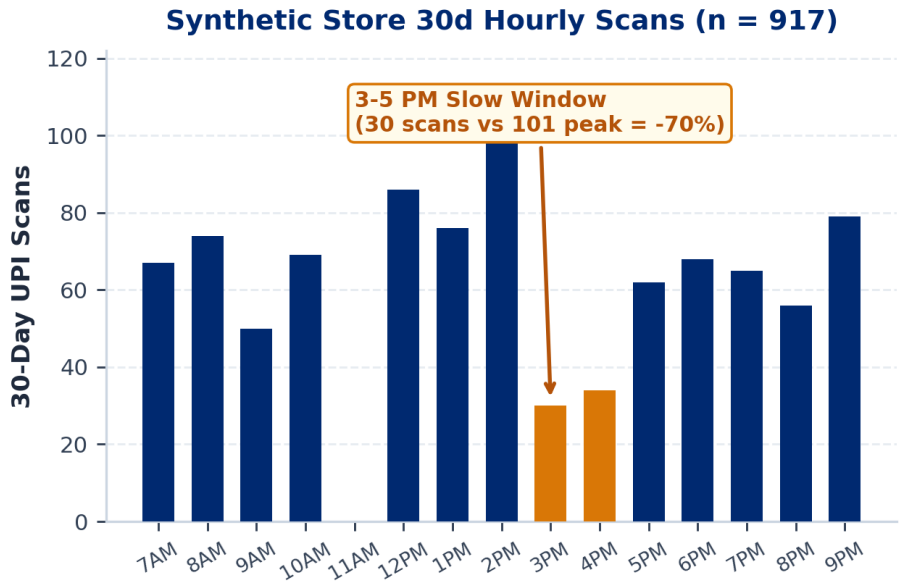
Merchants hear isolated payment alerts without customer-level retention tracking, losing repeat buyers before cashflow drops are noticed.

2. Dashboard Friction (<8% WAU - ASSUMPTION)

Counter-operated kirana, darshini, and pharmacy shops lack time to write prompts, export CSVs, or configure CRM rules.

3. Slow Hours & Rush Prep (SYNTHETIC STORE)

In our 60-day synthetic Indiranagar ledger, 3 to 5 PM scans fall 70% below peak while weekend morning rushes risk staple stockouts.



- **Reported (Paytm Q2 FY27, 26 Oct 2026):** 1.64 Cr subscription-paying merchants, Rs 7.1L Cr merchant GMV, Rs 2,648 Cr operating revenue.
- **Assumptions vs Synthetic:** ~7% merchant loan pen. & Rs 90-100/mo Soundbox fee (Assumptions); Ramesh Rao store is synthetic (917 scans/30d).

Track 1 Coverage: Growth, Operations, and Customer Recovery

PILLAR 1 : GROW BUSINESS

Retention & Slow-Hour Copilot

- **Churn Detection (Built):** SQL flags regulars silent 16+ days (16-38d in synthetic DB).
- **Slow-Hour Group (Built):** Identifies 3-5 PM snack buyers (55 detected, 34 eligible).
- **Cap & Stop-Loss (Built):** Cap Editor (Rs 50-2,500), 2-step voice confirm & Stop-Loss.
- **QR Coupon Hook (Planned):** Production hook into Paytm QR redemption rail.

PILLAR 2 : MANAGE OPERATIONS

Rush Prep & Credit Signals

- **Rush Alert (Simulated):** Flags weekend morning demand spikes from scan history.
- **Supplier PO (Simulated):** Dispatches pre-filled WhatsApp restock (+18L milk, +12kg batter).
- **Dual Ledgers (Built):** Separates merchant cashback stop from 30% Sarvam credit cap.
- **Credit Signal (Planned):** Feeds verified cashflow stability to partner NBFC funnel.

PILLAR 3 : SERVE CUSTOMERS

Privacy-Safe Service Recovery

- **Delay Tagging (Simulated):** Flags peak-hour queue delays and 3-star feedback tags.
- **Goodwill Pass (Simulated):** 1-tap bilingual apology + Rs 30 next-scan coffee voucher.
- **Zero PII Leakage (Built):** No raw phone numbers or payer IDs ever leave SQLite.
- **DPDP & Cooldown (Built):** Hard opt-in filter and 14-day per-payer anti-spam cooldown.

Intention-to-Treat Cap-vs-Holdout & Late-Arrival Handling

STEP 1 : PASS ISSUANCE

Filter 48 -> 40 eligible (80/20: 32 nudged, 8 control). Issue Rs 25 passes ('valid while shop offer budget lasts') to ALL 32 nudged buyers (Rs 800 face).

STEP 2 : >10% & 80% ALERTS

At approval, P(Binomial(32, 0.375) > 12) = 42.2% (>10% alert -> suggests Rs 400 cap). Live at 80% cap (10/12 paid), prompt merchant 1-tap top-up.

STEP 3 : INDEPENDENT DRAW

48h simulator draws customer return visits independently of max_budget first, then applies atomic BEGIN IMMEDIATE stop at Rs 300 (12 paid).

STEP 4 : ITT & DECLINE LOG

After cap, normal UPI clears, cashback withholds, refused_redemptions + audit_log increment (brand risk tracked). ITT (32 vs 8) stays unchanged.

DISCLOSED BRAND RISK & VERIFIED INVARIANCE TESTS

- **Post-Cap Customer Experience & Brand Risk:** Pass terms state 'valid while shop offer budget lasts'; declining a 13th regular still risks counter disappointment, so overrun alerts, 80%-cap top-up prompts, and logged declines are mandatory.
- **Automated Concurrency & ITT Tests:** 25-thread simultaneous scan test verifies budget_spent == Rs 300 (12 paid, 13 refused); multi-seed test proves ITT holdout comparison is identical whether cap is Rs 800 or Rs 100.

Component Status: Built vs Pending vs Production Plan

TOUCHPOINTS	[BUILT] Merchant & Technical View UI	[BUILT] Mic + Fallback Chips + Repeat/0.78x	[PLANNED] Soundbox 4G Push & WhatsApp Cloud
VOICE & LLM	[BUILT] Browser Speech + SHA-256 Cache	[PENDING LIVE] Keyless Sarvam (30% Cap)	[BUILT/OPT] Keyless Local Q&A Guard
CORE ENGINE	[BUILT] FastAPI + TTL/LRU/Byte Sessions	[BUILT] 6-Rule Guardrail Lab + Cap Editor	[BUILT] ITT + Bayesian + Sensitivity Engine
DATA & OPS	[BUILT] Atomic SQLite Cap & Stop-Loss	[BUILT] Preflight + Unit-Econ Generator	[PLANNED] PostgreSQL & Redis Cluster

Differentiation vs Manual Khata Apps and Generic Chatbots

Dimension	Manual BI / Khata Apps	Generic LLM Chatbots	PaytmAI VyaparVoice
Merchant UX	Charts & manual ledger entry	Requires typing business prompts	**8:30 PM plain voice + Repeat/0.78x + 1-tap**
Customer Graph	Incomplete manual phonebooks	No link to payment history	**Paytm QR repeat-payer IDs (synthetic in MVP)**
Financial Safety	Manual coupon rules	LLM can invent unsafe offers	**6-rule guardrail lock, Cap Editor & Stop-Loss**
Privacy (DPDP)	Raw phone lists exposed	Prompt data leakage risk	**SQL opt-in filter; zero PII sent to voice/LLM**
Causal Proof	Vanity message counts	No comparison group	**80/20 ITT + Bayesian posterior + 50-shop pilot**

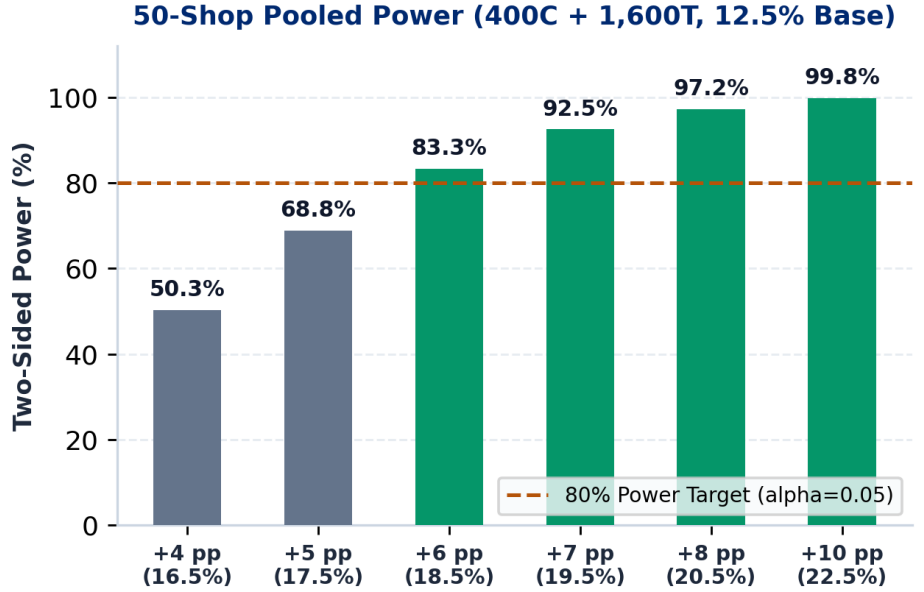
50-Shop Stratified Power (400C + 1,600T) & Pilot Targets

POOLED 50-SHOP X 1-WAVE POWER (12.5% BASELINE, ALPHA=0.05)

- **Why Pool Stores:** 50 shops x 1 wave (~400C + ~1,600T, stratified by shop); avoids claiming 1 shop runs 4 waves in 4w under a 14d cooldown (~0.78 waves/mo).
- **Exact Power Curve:** +4 pp: 50.3% | +5 pp: 68.8% | +6 pp: 83.3% | +7 pp: 92.5% | +8 pp: 97.2% | +10 pp: 99.8%.
- **Break-Even Lift:** +7.8 pp pre-fee (20.3%) and +15.3 pp post-5%-fee (27.8% at 30% shop margin; >99.9% power).

ECOSYSTEM HYPOTHESES (UNVALIDATED PILOT TARGETS)

- **Repeat & Waste Targets:** +14% to +22% repeat visits; -18% afternoon perishable loss (PILOT TARGETS).
- **Device & QR Share:** -35% Soundbox churn; +12% to +15% store Paytm QR GMV lift (PILOT TARGETS).
- **Credit Funnel:** Expand ~7% merchant loan pen. (ASSUMPTION) toward 15% (LONG-TERM TARGET).



83.3%

Power at +6 pp Lift (400C + 1,600T)

+15.3 pp

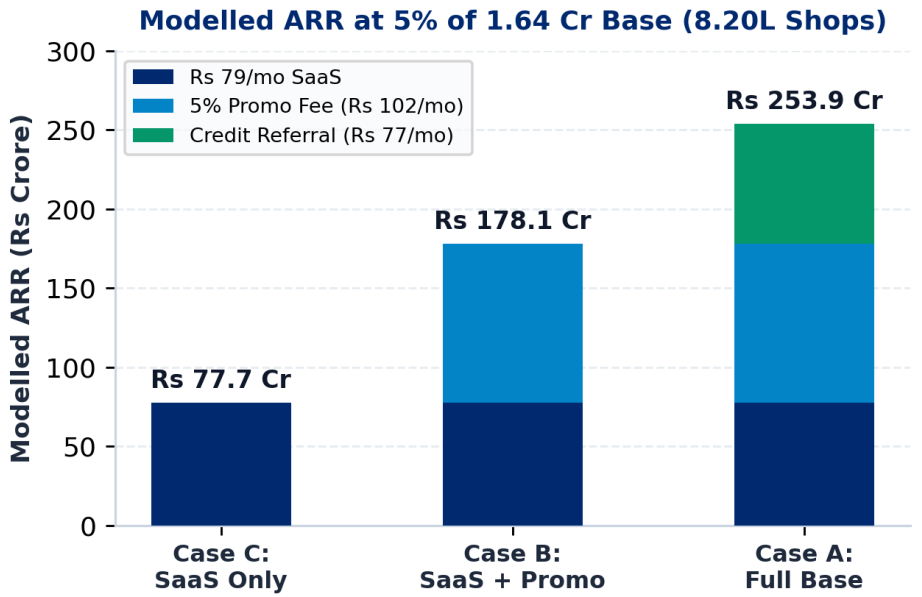
Post-Fee Break-Even (+7.8 pp Pre-Fee)

Rs 5.00

Cached AI Cost/Mo (Rs 28.80 Uncached)

Q2 FY27 ARR Cases, Fee Reconciliation & TTS Cache Sensitivity

- **Stream 1 (SaaS - Rs 79/mo):** Add-on to Soundbox (~Rs 90-100/mo rental ASSUMPTION) x 8.20L shops (5% of 1.64 Cr Q2 FY27 base) = Rs 77.7 Cr ARR.
- **Stream 2 (Promo Fee - Rs 102/mo):** 5% fee on ALL redeemed bills incl. organic returners: 1 wave = Rs 130.50 (Rs 2,610 bills); steady state assumes ~0.78 waves/mo (Rs 2,040/mo bills) = Rs 102/mo (+Rs 100.4 Cr -> Rs 178.1 Cr ARR).
- **Stream 3 (Credit - Rs 77/mo):** Amortized 1.5%-2.5% partner origination uplift (~7% loan pen. ASSUMPTION) -> +Rs 75.8 Cr (Rs 253.9 Cr ARR).
- **Cached COGS (Rs 48.00/mo):** Rs 1.80 SLM + Rs 3.20 TTS (~88% phrase cache hit on Bulbul v3) + Rs 43 msg = 81.4% gross margin.
- **Uncached COGS (Rs 71.80/mo):** Uncached Bulbul v3 (9k chars at Rs 30/10k = Rs 27 TTS) = 72.2% margin (39.2% cached margin on SaaS-only).



REPRODUCIBLE SCRIPT (scripts/unit_economics.py)

1. Fee Base: 5% applies to gross QR-redeemed promo bills including organic returners.

2. Cadence: Initial backlog wave yields Rs 130.50; steady replenishment (~0.78 waves/mo) yields Rs 102/mo.

3. Q2 FY27 Base: 5% x 1.64 Cr = 8.20L shops.

APPENDIX A9 : VALIDATION & GO / NO-GO DECISION RULE

From Verified Stage 2 MVP to Pooled 50-Shop Stratified Pilot

STAGE A : BUILT & VERIFIED

Deterministic Core & Stage 2 UX

- Bounded visitor SQLite sandboxes (204 KB/visitor; 55 MB peak RSS)
- Cap Editor (Rs 50-2,500), 2-step voice confirm, Stop-Loss & Guardrail Lab
- Bayesian profit posterior, 6x4 sensitivity matrix & 5-point teach-back

STAGE B : POOLED 50-SHOP PILOT

Stratified Single-Wave Validation

- 3-merchant counter teach-back across EN (author-checked), KN & HI (draft)
- 50 shops x 1 wave (~400C + ~1,600T, stratified by shop; 83.3% power at +6 pp)
- Advance only if lower 95% bound of post-5%-fee net profit/nudged > 0 AND lift > break-even (+15.3 pp)

STAGE C : PRODUCTION SCALE

Paytm Ecosystem Integration

- Bind Paytm QR dynamic coupon settlement & 4G Soundbox push
- Enforce live 80%-cap merchant top-up prompt to protect counter goodwill
- Scale across 8.20L target stores (Rs 77.7 Cr to Rs 253.9 Cr ARR)

Merchant Plain-Language View vs Technical Audit View

MERCHANT VIEW (ZERO JARGON - LINT VERIFIED)

- **Plain Verdict & Why?:** Promising single-shop check (+Rs 222 pre-fee, +Rs 91.50 post-fee); 1-tap Why? drawers explain SQL triggers in plain words.
- **Counter Ergonomics:** Repeat Last, 0.78x Slower mode, and 5 Merchant Question Chips documented in docs/TEACH_BACK.md.
- **Clear Spend Stop:** Your 300 rupee limit was reached after 12 paid redemptions (0 extra declined in base run).
- **Honest Copy Status:** EN author-checked, native review pending; KN & HI carry visible [DRAFT - UNREVIEWED] badges.

TECHNICAL & AUDIT VIEW (ANALYST / JUDGE)

- **ITT & Wilson 95% CIs:** 37.5% [22.9%, 54.8%] vs 12.5% [2.2%, 47.1%]; two-sided Fisher exact p = 0.24.
- **Bayesian Posterior:** Beta(1,1) uniform prior (10k draws): 88.8% P(lift>0), 67.0% P(pre-fee>0), 52.1% P(post-fee>0).
- **Guardrail Red-Team Lab:** Interactive 'Type your own attack' tests 6 hard rules (injection, <=22% discount, DPDP, 14d, cap, holdout).
- **Banned Merchant Terms:** 25 regex rules block cohort, holdout, p-value, Fisher, Wilson, ROI, DPDP, RFM, ITT.

GRADE A : OBSERVED LEDGER

SQLite store counts (240 payers, 1,613 scans, 48 silent -> 40 eligible; synthetic seed 20261025).

GRADE B : SIMULATED ITT

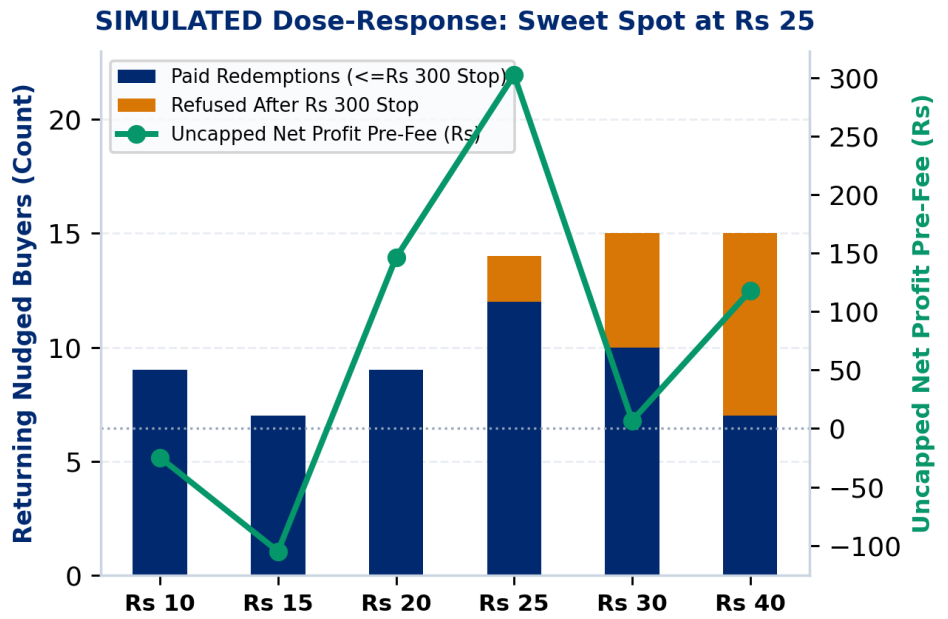
Randomized 32 vs 8 split, independent outcome draw, Fisher p & Wilson CIs (simulated).

GRADE C : ASSUMPTIONS

30% shop gross margin, 5% fee on redeemed bills, Beta(1,1) profit posterior & sensitivity table.

Merchant Spend Stop, Sarvam Credit Cap & Dose-Response Curve

- **Merchant Cashback Stop:** Issues passes ('valid while budget lasts') to all 32 nudged buyers; SQLite lock stops spend at Rs 300 (12 paid) and logs declines.
- **Dose-Response Sweet Spot:** In simulation, Rs 25 peaks uncapped profit: offers of Rs 30 to 40 erode uncapped net profit via organic subsidy AND trigger 5 to 8 cap refusals under a Rs 300 stop.
- **Margin & Lift Break-Even:** At +25 pp simulated lift, shop breaks even at 17.2% margin pre-fee (24.7% post-fee); at 30% margin, break-even lift is +7.8 pp pre-fee (+15.3 pp post-fee).
- **Sarvam 30% Credit Ledger:** Separate JSON ledger caps TTS <=15% and STT <=15% (30% total of 500 sponsor credits = 150 max) and halts fail-closed on HTTP 402/429/5xx.
- **Public Cloud Lock:** PUBLIC_DEPLOY=true (auto-enabled on Render) hard-blocks live paid calls while SHA-256 audio cache replays at 0 credits.



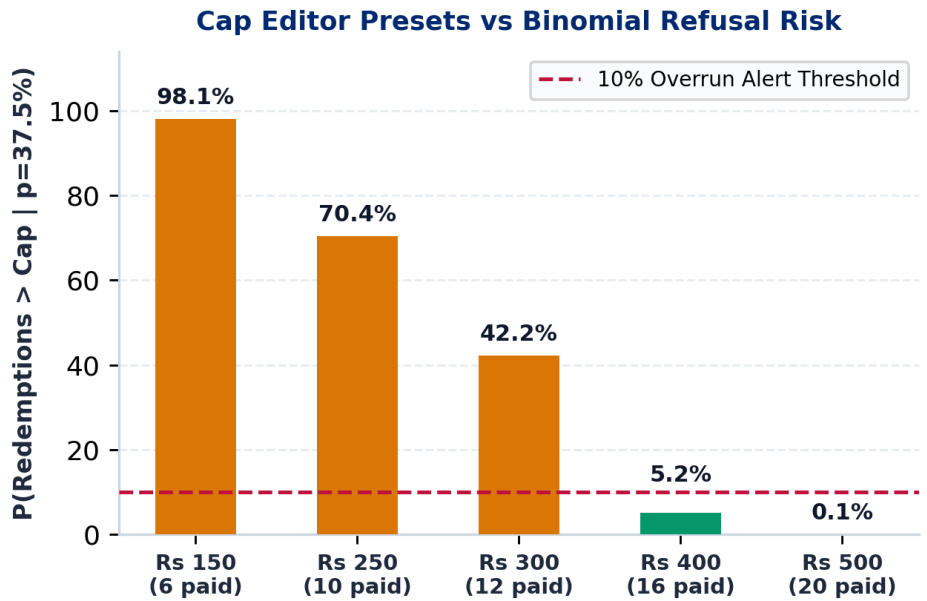
DUAL CAP SUMMARY (SIMULATED & BUILT)

1. Merchant Cap: Protects shopkeeper pocket (Rs 300 stop; >10% overrun warning & 80% top-up prompt).

2. Sponsor API Cap: Protects hackathon credits (<=15% TTS, <=15% STT, fail-closed).

Bounded Cap Editor, Two-Step Voice Confirm & Live Interim Stop-Loss

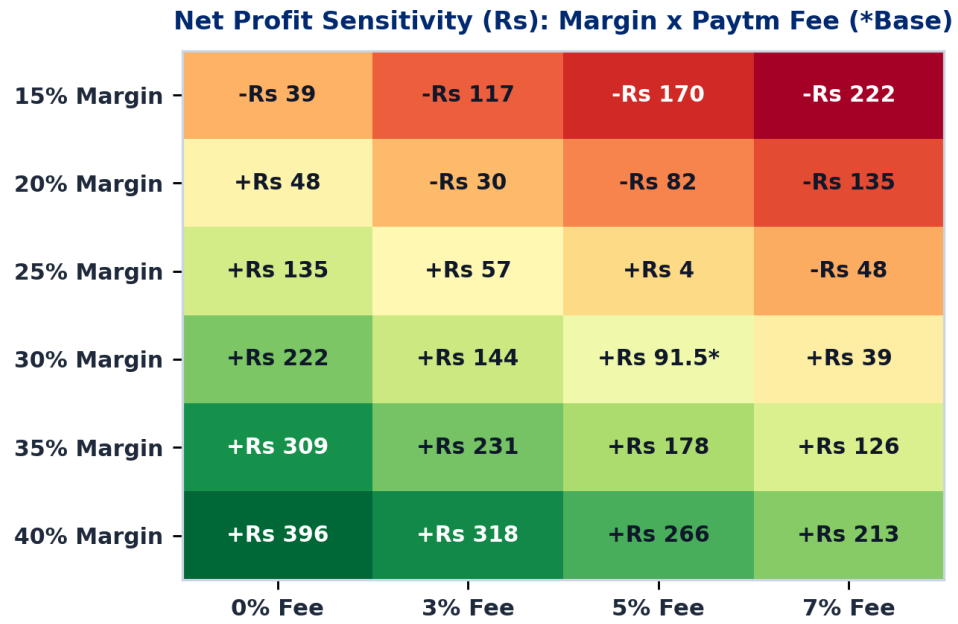
- **Bounded Cap Editor [BUILT]:** Enforces hard bounds [Rs 50, Rs 2,500] (>=1 pass of Rs 25) with 1-tap presets: Rs 150, Rs 250, Rs 300 (base), Rs 400 (<=10% overrun), Rs 500.
- **Two-Step Voice Confirmation [BUILT]:** Saying 'Set my limit to 400 rupees' stages CAP_CONFIRMATION_PENDING without mutating SQLite until merchant says 'Confirm limit' or taps Confirm.
- **Live Interim Wave Monitor [BUILT]:** Stepping live scans displays an explicit INTERIM DIRECTIONAL READOUT banner so shopkeepers never stop a wave early on noise.
- **Atomic 1-Tap Stop-Loss [BUILT]:** Triggering Stop-Loss sets PAUSED_STOP_LOSS and blocks pass redemptions inside BEGIN IMMEDIATE (BLOCKED_STOP_LOSS).
- **Immutable Audit Trail [BUILT]:** Every preset click, rejected out-of-bounds value, voice request/confirm, and Stop-Loss toggle writes to audit_logs.



CAP PRESETS VS BINOMIAL REFUSAL RISK (p = 37.5%, 32 PASSES)
Rs 150 (6 paid): 98.1% overrun | Rs 250 (10 paid): 70.4%
Rs 300 (12 paid): 42.2% (>10% alert) | Rs 400 (16 paid): 5.2% (SAFE)
Rs 500 (20 paid): 0.1% overrun.

Beta(1,1) Profit Posterior, Evidence Meter & 6x4 Margin-Fee Matrix

- **Plain-Language Verdict [BUILT]:** Merchant View translates single-wave output into plain rupee terms with zero statistical jargon, backed by Grade A/B/C badges.
- **Bayesian Profit Posterior [BUILT]:** Conjugate Beta(1,1) prior (10k draws, seed 20261025) without clamping negative draws: 88.8% P(lift>0), 67.0% P(pre-fee>0), 52.1% P(post-fee>0).
- **Why Single Wave Cannot Auto-Scale:** Single-wave 95% CrI is [-Rs 19.97, +Rs 14.73] per nudged buyer (due to n=8 control uncertainty), failing A9 scale gate until 50 shops are pooled.
- **4-Stage Visual Evidence Meter [BUILT]:** Tracks progress from n=40 (1 shop) -> n=155 (+25pp power) -> n=600 (15 shops) -> n=2,000 (50-shop stratified pilot, 83.3% power at +6 pp).
- **Kirana Margin Sensitivity [BUILT]:** At 5% fee, Rs 25 pass profits at >=25% margin (+Rs 4.50 at 25%, +Rs 91.50 at 30%) and loses at 15%-20% (-Rs 82.50 at 20%), guiding smaller kirana offers.



HONEST SCORECARD TAKEAWAY (TRACED TO unit_economics.md)

Base [30% Margin, 5% Fee] = +Rs 91.50 (+Rs 222 pre-fee).

Low-margin grocery [15%-20% Margin] requires lowering pass to Rs 10-15 on Rs 250+ bills (4%-6% discount).